A3-PS035-TC457

PS-035 : Evidence - Based Operator Network Fault Localisation

TRAZELY: Evidence-Based Diagnosis Pipeline

Rule-based network fault diagnosis using topology, syslog, and ping evidence.

Team Code Crafters

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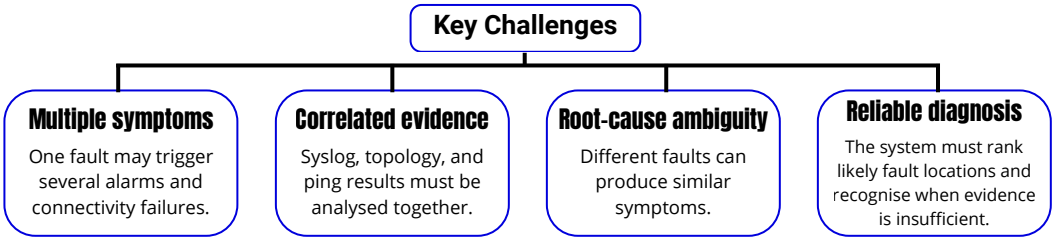
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Problem Description

Evidence-Based Operator Network Fault Localisation

Network failures can generate multiple symptoms across logs, connectivity checks, and network topology, making it difficult to identify the most likely root cause.



Project Objective

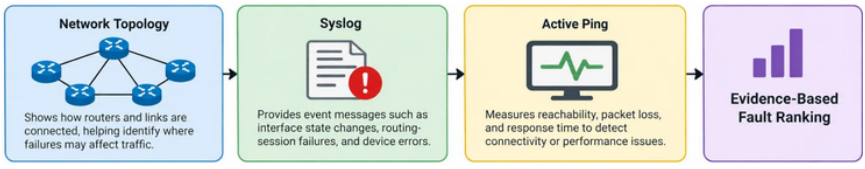
Develop a system that uses network evidence to rank likely faults and evaluate its localisation accuracy on held-out fault scenarios.

Background

Network Fault Localisation

Network fault localisation identifies the likely source of a failure by analysing observable network symptoms.

Evidence Sources



Proposed Optimisation

A single evidence source may be misleading. Combining topology, syslog, and ping evidence enables a rule-based engine to rank likely fault candidates, rather than relying on one symptom alone.

Evaluation Principle

Diagnosis quality will be measured using Top-1 and Top-3 accuracy on held-out fault runs, with insufficient or conflicting evidence explicitly recognised.

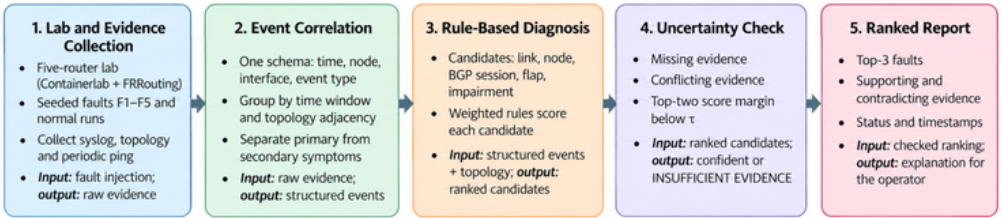
Literature & RFC Review

Research paper / standard	Existing contribution	Relevance to TRAZELY
Steinder & Sethi (2004) — A Survey of Fault Localization Techniques in Computer Networks	Reviews fault-localisation methods, event correlation, and root-cause analysis.	Motivates correlating multiple symptoms to rank likely faults.
IETF 126 Hackathon – AI-Assisted Network Fault Localization	Explores correlating network symptoms, telemetry, topology, configuration logs, and alarms to identify probable fault locations and supporting evidence.	Motivates multi-source evidence correlation. TRAZELY takes a narrower approach using topology, syslog, and active ping with a rule-based fault-ranking baseline.
RFC 5424 — The Syslog Protocol	Defines a standard message format for system event logging.	Supports consistent collection and interpretation of syslog evidence.
RFC 4271 — BGP-4	Defines the Border Gateway Protocol used to exchange routing information between autonomous systems.	Relevant if BGP-session failure is included in the lab's fault scenarios.

Existing work explores symptom correlation and probabilistic diagnosis; TRAZELY proposes a lightweight, explainable rule-based baseline using topology, syslog, and ping evidence.

Proposed Solution

TRAZELY: Evidence-Based Diagnosis Pipeline



Optional Random Forest

- Added only after the baseline works
- Trained on development runs, using evidence features only
- Compared on the same held-out runs
- Feeds into stage 3 as a second ranking, never replaces the evidence display

A working rule-based system comes first. Machine learning is an extension, not a dependency.

Optimization Proposed

Basic approach vs TRAZELY

Basic approach	TRAZELY improvement	Intended benefit
Alarms read one by one	Correlate by time and topology	Fewer secondary alarms mistaken for the root cause
Single best guess	Ranked top-3 with evidence	Operator can review alternatives
Forced diagnosis	Insufficient-evidence reporting	Abstains on weak or conflicting evidence
Logs and pings viewed separately	Combined logs, pings and topology	Separates look-alike faults (e.g. F1 vs F3)
Tuned on all data	Development and held-out split	Honest accuracy measurement

Benefits are intended. No accuracy gain has been demonstrated yet.

Solution Architecture & Design

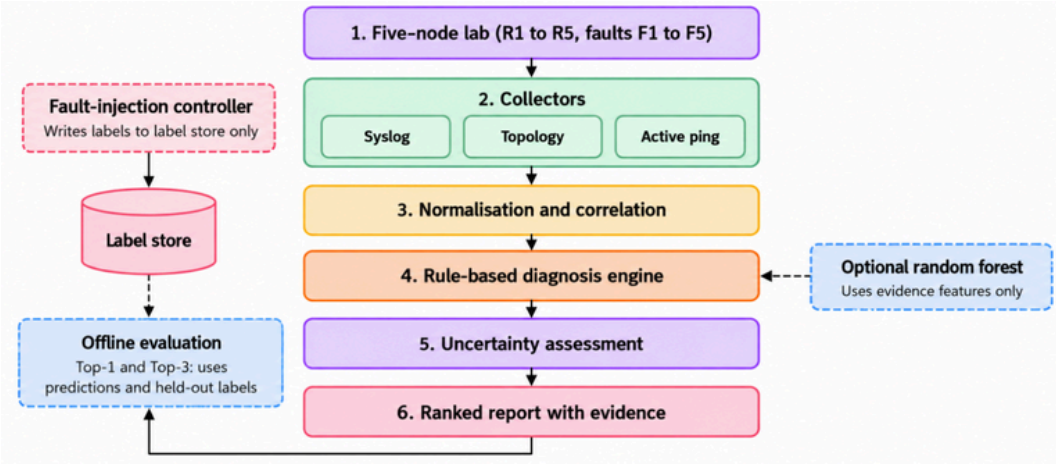


Figure: Proposed architecture. Ground-truth labels never reach the diagnosis stages.

Data Flow Diagram

Proposed data movement between processes, stores and entities

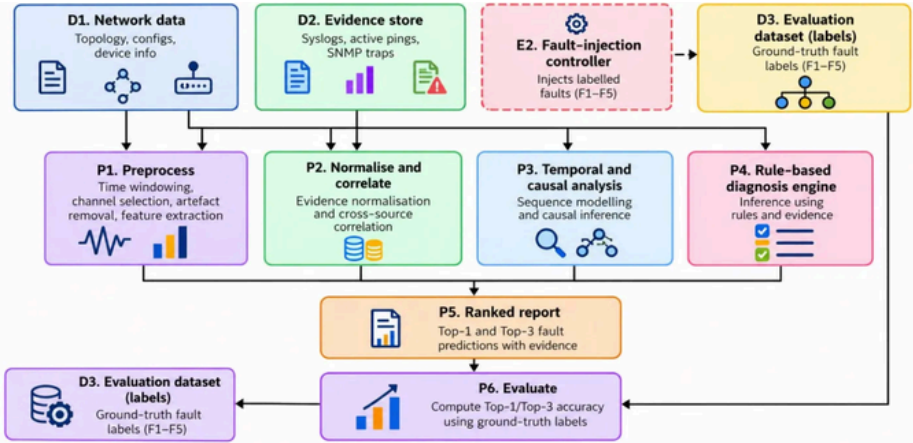


Figure: Proposed data flow. No flow leads from diagnosis to device configuration.

Deployment Timeline

Planned phases: baseline first, optional ML last

Phase 0: Feasibility

Work:
- 3 routers, one BGP session, one fault, timed reset.

Completion evidence:
- Working test run.

Phase 1: Five-node lab

Work:
- Topology and five resettable faults.

Completion evidence:
- Diagram and reset procedure.

Phase 2: Evidence

Work:
- Syslog, topology, ping, plus normal runs.

Completion evidence:
- Raw logs and measurements.

Phase 3: Baseline

Work:
- Rule-based ranking with evidence.

Completion evidence:
- Ranked output.

Phase 4: Evaluation

Work:
- Held-out runs, Top-1/Top-3, unsupported cases.

Completion evidence:
- Metrics table.

Phase 5: Optional ML

Work:
- Random forest vs baseline.

Completion evidence:
- Comparison table.

Phase 6: Final demo

Work:
- Report PDF, demo, pseudocode.

Completion evidence:
- Repository and demo.

Bibliography

[1] AIORI-3, “PS-035: Evidence-Based Operator Network Fault Localisation,” problem statement.
[2] IETF 126 Hackathon, “AI-Assisted Network Fault Localization” (linked in PS-035).
[3] M. Steinder and A. Sethi, “A survey of fault localization techniques in computer networks,” Science of Computer Programming, 53(2), 2004.
[4] RFC 5424, “The Syslog Protocol.”
[5] RFC 4271, “A Border Gateway Protocol 4 (BGP-4).”
[6] FRRouting, Containerlab and tc-netem documentation.

THANK YOU!

TRAZELY: Evidence-Based Diagnosis Pipeline

Team : A3-PS035-TC457

• AIORI Hackathon 3

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